

Thang Duong

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SUMMARY

PhD candidate and NeurIPS 2024 first author specializing in LLM reasoning and Reinforcement Learning. Seeking to apply 5+ years of research experience from academia and industry (Qualcomm) to solve LLM reasoning and large-scale recommendation & personalization challenges. Proven record of publishing top-tier work, with up to 3x faster RL convergence (in our experiments) and a 2x improvement in wireless beam alignment benchmarks. Passionate about LLM reasoning and sample-efficient RL at scale.

EDUCATION

The University of Arizona

Tucson, AZ

PhD. Candidate (expected Dec. 2026) in Computer Science

Aug. 2022 – Present

Research focus: High-dimensional Interactive Learning by utilizing domain specific inductive biases (Advisor: Prof. Chicheng Zhang).

Hanoi University of Science and Technology

Hanoi, Vietnam

B.S.E. in Mechatronics Engineering, Advanced Program

Sep. 2012 – May 2018

PUBLICATIONS

NeurIPS 2024: *Beyond task diversity: Provable representation transfer for sequential multi-task linear bandits* – First author.

WiOpt 2026: *Physics-Informed Parametric Bandits for Beam Alignment in mmWave Communications* – Co-first author.

Working papers (in revision): *Efficient Sequential Multitask Bandit; Sample-efficient Reinforcement Learning by Warm-starting with LLMs.*

[Google Scholar Profile]

EXPERIENCE

Graduate Research Assistant

Aug. 2022 – Present

The University of Arizona

Tucson, AZ

- Provided a regret guarantee and eliminated the task diversity assumption to show that **Sequential Multitask Representation Transfer in Bandit** is applicable to real-world problems (**NeurIPS 2024**).
- Achieved **2x reduction** in beam alignment regret on DeepMIMO and DeepSense6G benchmarks by developing **physics-informed bandit algorithms** for Wireless Communication. Cross-team collaborated with Prof. Ming Li's lab from the ECE department.
- Built a novel **RL warm-start pipeline using LLM-collected demonstrations**, reaching **up to 3x faster** convergence on OpenAI Gym in our experiments. Mentoring one undergraduate. Enabled reproducible research by developing a modular evaluation suite and release scripts for large-scale experiments on H100 GPUs.

Visiting Student

May 2025 – Aug. 2025

Toyota Technological Institute at Chicago (TTIC)

Chicago, IL

- **RL Advantage analysis for LLMs' PRM:** Investigating the theoretical underpinnings of Process Reward Models (PRMs) in LLM reasoning by framing them within the Actor-Critic framework. This ongoing research, in collaboration with Prof. Chicheng Zhang et al., aims to provably improve the reasoning capabilities of LLMs as a PRM-guided search-based policy.

Research Resident

Dec. 2019 – June 2022

Qualcomm (formerly VinAI Research)

Hanoi, Vietnam

- **Bandits meta-learning:** Supported a comprehensive literature review and implemented a modular meta-learning codebase for multi-task bandits that leverages shared structure. Demonstrated improved sample-efficiency in a comprehensive suite of experiments and an ablation study on a synthetic dataset; results accepted at **RLC 2024**.
- **Active Learning for Domain Adaptation:** Prototyped several active learning strategies to explore their viability for domain adaptation and model warm-starting.

- **Sim-to-Real Data Augmentation:** Investigated Sim-to-Real transfer techniques by developing and testing a proof-of-concept domain adaptation method in the CARLA simulator.

Junior Engineer

July 2019 – Dec. 2019

Qualcomm (formerly VinAI Research)

Hanoi, Vietnam

- Engineered an Android application using OpenGL 2.0 to provide 3D visualization of facial reconstruction models, serving as the primary technology demonstration at the NeurIPS 2019 conference.

AI Team leader, Scrum master

May 2018 – June 2019

NAL Vietnam JSC

Hanoi, Vietnam

- Collaborated with the product and engineering teams to integrate multiple NLP models (e.g., intent classification, entity recognition) into Chatops, a commercial chat interface for business.
- Led a team of six members in the successful integration of Facial recognition models for a parent-child matching product, deployed in six kindergarten locations.

SKILLS

Core ML/AI: PyTorch, TensorFlow, HuggingFace, Transformers, OpenAI Gym, RLlib, Claude Code, Codex, Agentic Engineering

Languages: Python, Java, C/C++

Developer Tools: Git, Docker, Linux, CUDA

Research: Online Learning, Reinforcement Learning, Multi-task Learning, Bandit Theory, LLMs, Representation Transfer, Online PCA, LLM-RL alignment, Representation Learning, Meta-learning, Machine Learning, Statistics, Mathematics, Deep learning, Foundational Models, Rapid Prototyping, data science